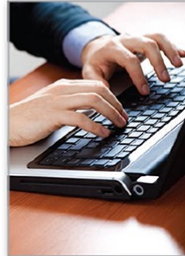
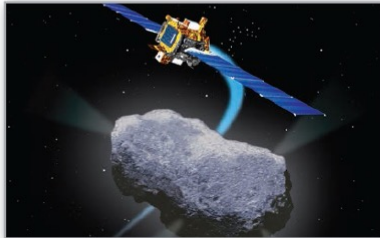


3 Determinants



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1

3.1 The Determinant of a Matrix

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Objectives

- Find the determinant of a 2×2 matrix.
- Find the minors and cofactors of a matrix.
- Use expansion by cofactors to find the determinant of a matrix.
- Find the determinant of a triangular matrix.

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The determinant of a 2×2 matrix

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The determinant of a 2×2 matrix (1 of 1)

Definition of the determinant of a 2×2 matrix

The **determinant** of the matrix

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

is $\det(A) = |A| = a_{11}a_{22} - a_{21}a_{12}$.

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Example 1 – Determinants of Matrices of Order 2

a. For $A = \begin{bmatrix} 2 & -3 \\ 1 & 2 \end{bmatrix}$, $|A| = \begin{vmatrix} 2 & -3 \\ 1 & 2 \end{vmatrix} = 2(2) - 1(-3) = 4 + 3 = 7$.

b. For $B = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$, $|B| = \begin{vmatrix} 2 & 1 \\ 4 & 2 \end{vmatrix} = 2(2) - 4(1) = 4 - 4 = 0$.

c. For $C = \begin{bmatrix} 0 & \frac{3}{2} \\ 2 & 4 \end{bmatrix}$, $|C| = \begin{vmatrix} 0 & \frac{3}{2} \\ 2 & 4 \end{vmatrix} = 0(4) - 2(\frac{3}{2}) = 0 - 3 = -3$.

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Minors and Cofactors

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Minors and Cofactors (1 of 1)

Minors and Cofactors of a Square Matrix

If A is a square matrix, then the **minor** M_{ij} of the entry a_{ij} is the determinant of the matrix obtained by deleting the i th row and j th column of A . The **cofactor** C_{ij} of the entry a_{ij} is $C_{ij} = (-1)^{i+j}M_{ij}$.

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Example 2 – Minors and Cofactors of a Matrix

Find all the minors and cofactors of

$$A = \begin{bmatrix} 0 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & 0 & 1 \end{bmatrix}.$$

Solution:

To find the minor M_{11} , delete the first row and first column of A and evaluate the determinant of the resulting matrix.

$$\begin{bmatrix} 0 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & 0 & 1 \end{bmatrix}, \quad M_{11} = \begin{vmatrix} -1 & 2 \\ 0 & 1 \end{vmatrix} = -1(1) - 0(2) = -1$$

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Example 2 – Solution

Verify that the minors are

$$M_{11} = -1 \quad M_{12} = -5 \quad M_{13} = 4$$

$$M_{21} = 2 \quad M_{22} = -4 \quad M_{23} = -8$$

$$M_{31} = 5 \quad M_{32} = -3 \quad M_{33} = -6$$

Now, to find the cofactors, combine these minors with the checkerboard pattern of signs for a 3×3 matrix shown above.

$$C_{11} = -1 \quad C_{12} = 5 \quad C_{13} = 4$$

$$C_{21} = -2 \quad C_{22} = -4 \quad C_{23} = 8$$

$$C_{31} = 5 \quad C_{32} = 3 \quad C_{33} = -6$$

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The Determinant of a Square Matrix

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The Determinant of a Square Matrix (1 of 2)

Definition of the Determinant of a Square Matrix

If A is a square matrix of order $n \geq 2$, then the determinant of A is the sum of the entries in the first row of A multiplied by their respective cofactors. That is,

$$\det(A) = |A| = \sum_{j=1}^n a_{1j}C_{1j} = a_{11}C_{11} + a_{12}C_{12} + \cdots + a_{1n}C_{1n}.$$

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The Determinant of a Square Matrix (2 of 2)

Theorem 3.1 Expansion by Cofactors

Let A be a square matrix of order n . Then the determinant of A is

$$\det(A) = |A| = \sum_{j=1}^n a_{ij}C_{ij} = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in} \quad \text{\textit{ith row expansion}}$$

or

$$\det(A) = |A| = \sum_{i=1}^n a_{ij}C_{ij} = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}. \quad \text{\textit{jth column expansion}}$$

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Example 4 – The Determinant of a Matrix of Order 4

Find the determinant of

$$A = \begin{bmatrix} 1 & -2 & 3 & 0 \\ -1 & 1 & 0 & 2 \\ 0 & 2 & 0 & 3 \\ 3 & 4 & 0 & -2 \end{bmatrix}.$$

Solution:

Notice that three of the entries in the third column are zeros. So, to eliminate some of the work in the expansion, use the third column.

$$|A| = 3(C_{13}) + 0(C_{23}) + 0(C_{33}) + 0(C_{43})$$

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Example 4 – Solution (1 of 2)

The cofactors C_{23} , C_{33} , and C_{43} have zero coefficients, so you need only find the cofactor C_{13} . To do this, delete the first row and third column of A and evaluate the determinant of the resulting matrix.

$$C_{13} = (-1)^{1+3} \begin{vmatrix} -1 & 1 & 2 \\ 0 & 2 & 3 \\ 3 & 4 & -2 \end{vmatrix} \quad \text{Delete 1st row and 3rd column.}$$

$$= \begin{vmatrix} -1 & 1 & 2 \\ 0 & 2 & 3 \\ 3 & 4 & -2 \end{vmatrix} \quad \text{Simplify.}$$

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Example 4 – Solution (2 of 2)

Expanding by cofactors in the second row yields

$$C_{13} = (0)(-1)^{2+1} \begin{vmatrix} 1 & 2 \\ 4 & -2 \end{vmatrix} + (2)(-1)^{2+2} \begin{vmatrix} -1 & 2 \\ 3 & -2 \end{vmatrix} + (3)(-1)^{2+3} \begin{vmatrix} -1 & 1 \\ 3 & 4 \end{vmatrix}$$

$$= 0 + 2(1)(-4) + 3(-1)(-7)$$

$$= 13.$$

You obtain

$$|A| = 3(13)$$

$$= 39.$$

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Triangular Matrices

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Triangular Matrices (1 of 2)

A square matrix is *upper triangular* when it has all zero entries below its main diagonal, and *lower triangular* when it has all zero entries above its main diagonal, as shown in the diagram at the left.

A matrix that is both upper and lower triangular is a **diagonal matrix**. That is, a diagonal matrix is one in which all entries above and below the main diagonal are zero.

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Triangular Matrices (2 of 2)

Theorem 3.2 Determinant of a Triangular Matrix

If A is a triangular matrix of order n , then its determinant is the product of the entries on the main diagonal. That is,

$$\det(A) = |A| = a_{11}a_{22}a_{33} \cdots a_{nn}.$$

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Example 6 – The Determinant of a Triangular Matrix

The determinant of the lower triangular matrix

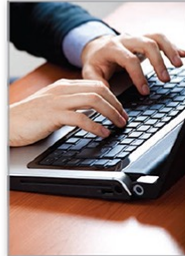
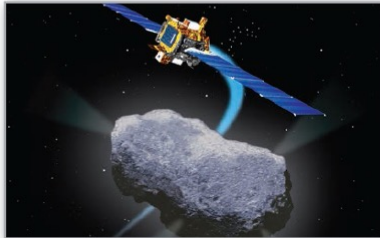
$$A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 4 & -2 & 0 & 0 \\ -5 & 6 & 1 & 0 \\ 1 & 5 & 3 & 3 \end{bmatrix}$$

is $|A| = (2)(-2)(1)(3) = -12$.

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3 Determinants



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3.2 Determinants and Elementary Operations

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Objectives

- Use elementary row operations to evaluate a determinant.
- Use elementary column operations to evaluate a determinant.
- Recognize conditions that yield zero determinants.

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Determinants and Elementary Row Operations

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Determinants and Elementary Row Operations (1 of 1)

Theorem 3.3 Elementary Row Operations and Determinants

Let A and B be square matrices.

1. When B is obtained from A by interchanging two rows of A , $\det(B) = -\det(A)$.
2. When B is obtained from A by adding a multiple of a row of A to another row of A , $\det(B) = \det(A)$.
3. When B is obtained from A by multiplying a row of A by a nonzero constant c , $\det(B) = c \det(A)$.

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Example 2 – Finding a Determinant Using Elementary Row Operations

Find the determinant of

$$A = \begin{bmatrix} 0 & -7 & 14 \\ 1 & 2 & -2 \\ 0 & 3 & -8 \end{bmatrix}.$$

Solution:

Using elementary row operations, rewrite A in triangular form as shown below.

$$\begin{vmatrix} 0 & -7 & 14 \\ 1 & 2 & -2 \\ 0 & 3 & -8 \end{vmatrix} = - \begin{vmatrix} 1 & 2 & -2 \\ 0 & -7 & 14 \\ 0 & 3 & -8 \end{vmatrix} \quad \begin{matrix} \leftarrow \\ \leftarrow \end{matrix} \quad \text{Interchange the first two rows.}$$

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Example 2 – Solution

$$= 7 \begin{vmatrix} 1 & 2 & -2 \\ 0 & 1 & -2 \\ 0 & 3 & -8 \end{vmatrix} \quad \leftarrow \text{Factor } -7 \text{ out of the second row.}$$

$$= 7 \begin{vmatrix} 1 & 2 & -2 \\ 0 & 1 & -2 \\ 0 & 0 & -2 \end{vmatrix} \quad \leftarrow \text{Add } -3 \text{ times the second row to the third row to produce a new third row.}$$

The above matrix is triangular, so the determinant is

$$|A| = 7(1)(1)(-2) = -14.$$

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Determinants and Elementary Column Operations

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Determinants and Elementary Column Operations (1 of 1)

Operations performed on the columns (rather than on the rows) of a matrix are **elementary column operations**, and two matrices are **column-equivalent** when one can be obtained from the other by elementary column operations.

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Example 3 – Finding a Determinant Using Elementary Column Operations

Find the determinant of

$$A = \begin{bmatrix} -1 & 2 & 2 \\ 3 & -6 & 4 \\ 5 & -10 & -3 \end{bmatrix}.$$

Solution:

The first two columns of A are multiples of each other, so you can obtain a column of zeros by adding 2 times the first column to the second column, as shown below.

$$\begin{vmatrix} -1 & 2 & 2 \\ 3 & -6 & 4 \\ 5 & -10 & -3 \end{vmatrix} = \begin{vmatrix} -1 & 0 & 2 \\ 3 & 0 & 4 \\ 5 & 0 & -3 \end{vmatrix}$$

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Example 3 – Solution

At this point, you do not need to rewrite the matrix in triangular form, because there is an entire column of zeros. Simply conclude that the determinant is zero. Specifically, by expanding by cofactors in the second column, you have

$$|A| = (0)C_{12} + (0)C_{22} + (0)C_{32} = 0.$$

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Matrices and Zero Determinants

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Matrices and Zero Determinants (1 of 1)

Theorem 3.4 Conditions That Yield a Zero Determinant

If A is a square matrix and any one of the conditions below is true, then $\det(A) = 0$.

1. An entire row (or an entire column) consists of zeros.
2. Two rows (or columns) are equal.
3. One row (or column) is a multiple of another row (or column).

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Example 4 – A Matrix with a Zero Determinant

Find the determinant of $A = \begin{bmatrix} 1 & 4 & 1 \\ 2 & -1 & 0 \\ 0 & 18 & 4 \end{bmatrix}$.

Solution:

Adding -2 times the first row to the second row produces

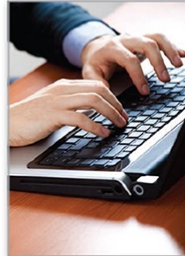
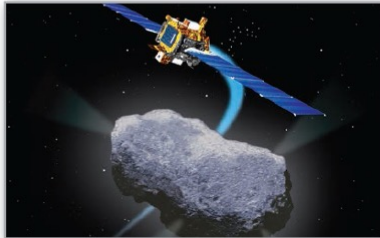
$$\begin{aligned} |A| &= \begin{vmatrix} 1 & 4 & 1 \\ 2 + (-2)(1) & -1 + (-2)(4) & 0 + (-2)(1) \\ 0 & 18 & 4 \end{vmatrix} \\ &= \begin{vmatrix} 1 & 4 & 1 \\ 0 & -9 & -2 \\ 0 & 18 & 4 \end{vmatrix}. \end{aligned}$$

The second and third rows are multiples of each other, so the determinant is zero.

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3 Determinants



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3.3 Properties of Determinants

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Objectives

- Find the determinant of a matrix product and a scalar multiple of a matrix.
- Find the determinant of an inverse matrix and recognize equivalent conditions for a nonsingular matrix.
- Find the determinant of the transpose of a matrix.

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Matrix Products and Scalar Multiples

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Example 1 – The Determinant of a Matrix Product

Find $|A|$, $|B|$, and $|AB|$ for the matrices

$$A = \begin{bmatrix} 1 & -2 & 2 \\ 0 & 3 & 2 \\ 1 & 0 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 2 & 0 & 1 \\ 0 & -1 & -2 \\ 3 & 1 & -2 \end{bmatrix}.$$

Solution:

$|A|$ and $|B|$ have the values

$$|A| = \begin{vmatrix} 1 & -2 & 2 \\ 0 & 3 & 2 \\ 1 & 0 & 1 \end{vmatrix} = -7 \quad \text{and} \quad |B| = \begin{vmatrix} 2 & 0 & 1 \\ 0 & -1 & -2 \\ 3 & 1 & -2 \end{vmatrix} = 11.$$

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Example 1 – Solution

The matrix product AB is

$$AB = \begin{bmatrix} 1 & -2 & 2 \\ 0 & 3 & 2 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 1 \\ 0 & -1 & -2 \\ 3 & 1 & -2 \end{bmatrix} = \begin{bmatrix} 8 & 4 & 1 \\ 6 & -1 & -10 \\ 5 & 1 & -1 \end{bmatrix}.$$

Finally,

$$|AB| = \begin{vmatrix} 8 & 4 & 1 \\ 6 & -1 & -10 \\ 5 & 1 & -1 \end{vmatrix} = -77.$$

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Matrix Products and Scalar Multiples (1 of 1)

Theorem 3.5 Determinant of a Matrix Product

If A and B are square matrices of order n , then $\det(AB) = \det(A) \det(B)$.

Theorem 3.6 Determinant of a Scalar Multiple of a Matrix

If A is a square matrix of order n and c is a scalar, then the determinant of cA is

$$\det(cA) = c^n \det(A).$$

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Example 2 – The Determinant of a Scalar Multiple of a Matrix

Find the determinant of the matrix.

$$A = \begin{bmatrix} 10 & -20 & 40 \\ 30 & 0 & 50 \\ -20 & -30 & 10 \end{bmatrix}$$

Solution:

$$A = 10 \begin{bmatrix} 1 & -2 & 4 \\ 3 & 0 & 5 \\ -2 & -3 & 1 \end{bmatrix} \quad \text{and} \quad \begin{vmatrix} 1 & -2 & 4 \\ 3 & 0 & 5 \\ -2 & -3 & 1 \end{vmatrix} = 5$$

so apply Theorem 3.6 to conclude that

$$|A| = 10^3 \begin{vmatrix} 1 & -2 & 4 \\ 3 & 0 & 5 \\ -2 & -3 & 1 \end{vmatrix} = 1000(5) = 5000.$$

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Determinants and the Inverse of a Matrix

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Determinants and the Inverse of a Matrix (1 of 2)

Theorem 3.7 Determinant of an Invertible Matrix

A square matrix A is invertible (nonsingular) if and only if $\det(A) \neq 0$.

Theorem 3.8 Determinant of an Inverse Matrix

If A is an $n \times n$ invertible matrix, then $\det(A^{-1}) = \frac{1}{\det(A)}$.

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Example 4 – The Determinant of the Inverse of a Matrix

Find $|A^{-1}|$ for the matrix $A = \begin{bmatrix} 1 & 0 & 3 \\ 0 & -1 & 2 \\ 2 & 1 & 0 \end{bmatrix}$.

Solution:

One way to solve this problem is to find A^{-1} and then evaluate its determinant. It is easier, however, to apply Theorem 3.8 as shown below. Find the determinant of A ,

$$|A| = \begin{vmatrix} 1 & 0 & 3 \\ 0 & -1 & 2 \\ 2 & 1 & 0 \end{vmatrix} = 4$$

and then use the formula $|A^{-1}| = 1/|A|$ to conclude that $|A^{-1}| = \frac{1}{4}$.

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Determinants and the Inverse of a Matrix (2 of 2)

Equivalent Conditions for a Nonsingular Matrix

If A is an $n \times n$ matrix, then the statements below are equivalent.

1. A is invertible.
2. $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $n \times 1$ column matrix $\mathbf{b}$.
3. $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
4. A is row-equivalent to I_n .
5. A can be written as the product of elementary matrices.
6. $\det(A) \neq 0$

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Determinants and the Transpose of a Matrix

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Determinants and the Transpose of a Matrix (1 of 1)

Theorem 3.9 Determinant of a Transpose

If A is a square matrix, then

$$\det(A) = \det(A^T).$$

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Example 6 – The Determinant of a transpose

Show that $|A| = |A^T|$ for the matrix below.

$$A = \begin{bmatrix} 3 & 1 & -2 \\ 2 & 0 & 0 \\ -4 & -1 & 5 \end{bmatrix}$$

Solution:

To find the determinant of A , expand by cofactors in the second *row* to obtain

$$\begin{aligned} |A| &= 2(-1)^3 \begin{vmatrix} 1 & -2 \\ -1 & 5 \end{vmatrix} \\ &= (2)(-1)(3) \\ &= -6. \end{aligned}$$

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Example 6 – Solution

To find the determinant of

$$A^T = \begin{bmatrix} 3 & 2 & -4 \\ 1 & 0 & -1 \\ -2 & 0 & 5 \end{bmatrix}$$

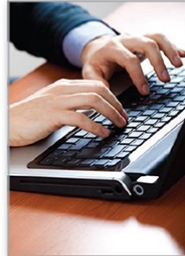
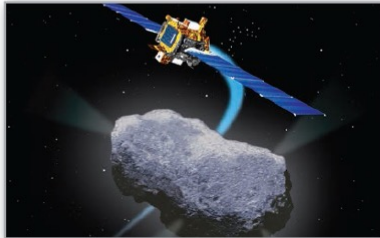
expand by cofactors in the second *column* to obtain

$$\begin{aligned} |A^T| &= 2(-1)^3 \begin{vmatrix} 1 & -1 \\ -2 & 5 \end{vmatrix} \\ &= (2)(-1)(3) \\ &= -6. \end{aligned}$$

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3 Determinants



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3.4 Applications of Determinants

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Objectives

- Find the adjoint of a matrix and use it to find the inverse of the matrix.
- Use Cramer's Rule to solve a system of n linear equations in n variables.
- Use determinants to find area, volume, and the equations of lines and planes.

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The Adjoint of a Matrix

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The Adjoint of a Matrix (1 of 3)

The cofactor C_{ij} of a square matrix A is $(-1)^{i+j}$ times the determinant of the matrix obtained by deleting the i th row and j th column of A .

The **matrix of cofactors** of A has the form

$$\begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & & \vdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{bmatrix}.$$

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The Adjoint of a Matrix (2 of 3)

The transpose of this matrix is the **adjoint** of A and is denoted $\text{adj}(A)$.

That is,

$$\text{adj}(A) = \begin{bmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{bmatrix}.$$

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Example 1 – Finding the Adjoint of a Square Matrix

Find the adjoint of $A = \begin{bmatrix} -1 & 3 & 2 \\ 0 & -2 & 1 \\ 1 & 0 & -2 \end{bmatrix}$.

Solution:

The cofactor C_{11} is

$$\begin{bmatrix} \boxed{-1} & 3 & 2 \\ 0 & -2 & 1 \\ 1 & 0 & -2 \end{bmatrix} \rightarrow C_{11} = (-1)^2 \begin{vmatrix} -2 & 1 \\ 0 & -2 \end{vmatrix} = 4.$$

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Example 1 – Solution

Continuing this process produces the matrix of cofactors of A shown below.

$$\begin{bmatrix} 4 & 1 & 2 \\ 6 & 0 & 3 \\ 7 & 1 & 2 \end{bmatrix}$$

The transpose of this matrix is the adjoint of A .

$$\text{That is, } \text{adj}(A) = \begin{bmatrix} 4 & 6 & 7 \\ 1 & 0 & 1 \\ 2 & 3 & 2 \end{bmatrix}.$$

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The Adjoint of a Matrix (3 of 3)

Theorem 3.10 The Inverse of a Matrix Using Its Adjoint

If A is an $n \times n$ invertible matrix, then $A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$.

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Example 2 – Using the Adjoint of a Matrix to Find Its Inverse

Use the adjoint of A to find A^{-1} ,

$$\text{where } A = \begin{bmatrix} -1 & 3 & 2 \\ 0 & -2 & 1 \\ 1 & 0 & -2 \end{bmatrix}.$$

Solution:

The determinant of this matrix is 3.

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Example 2 – Solution

Using the adjoint of A , the inverse of A is

$$A^{-1} = \frac{1}{|A|} \text{adj}(A) = \frac{1}{3} \begin{bmatrix} 4 & 6 & 7 \\ 1 & 0 & 1 \\ 2 & 3 & 2 \end{bmatrix} = \begin{bmatrix} \frac{4}{3} & 2 & \frac{7}{3} \\ \frac{1}{3} & 0 & \frac{1}{3} \\ \frac{2}{3} & 1 & \frac{2}{3} \end{bmatrix}.$$

Check that this matrix is the inverse of A by showing that

$$AA^{-1} = I = A^{-1}A.$$

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Cramer's Rule

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Cramer's Rule (1 of 1)

Theorem 3.11 Cramer's rule

If a system of n linear equations in n variables has a coefficient matrix A with a nonzero determinant $|A|$, then the solution of the system is

$$x_1 = \frac{\det(A_1)}{\det(A)}, \quad x_2 = \frac{\det(A_2)}{\det(A)}, \quad \dots, \quad x_n = \frac{\det(A_n)}{\det(A)}$$

where the i th column of A_i is the column of constants in the system of equations.

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Example 4 – Using Cramer's Rule

Use Cramer's Rule to solve the system of linear equations for x .

$$-x + 2y - 3z = 1$$

$$2x \quad \quad + \quad z = 0$$

$$3x - 4y + 4z = 2$$

Solution:

The determinant of the coefficient matrix is

$$|A| = \begin{vmatrix} -1 & 2 & -3 \\ 2 & 0 & 1 \\ 3 & -4 & 4 \end{vmatrix} = 10.$$

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Example 4 – Solution

The determinant is nonzero, so you know that the solution is unique. Apply Cramer's Rule to solve for x , as shown below.

$$x = \frac{\begin{vmatrix} 1 & 2 & -3 \\ 0 & 0 & 1 \\ 2 & -4 & 4 \end{vmatrix}}{10} = \frac{(1)(-1)^5 \begin{vmatrix} 1 & 2 \\ 2 & -4 \end{vmatrix}}{10} \\ = \frac{(1)(-1)(-8)}{10} = \frac{4}{5}$$

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Area, Volume, and Equations of Lines and Planes

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Area, Volume, and Equations of Lines and Planes (1 of 6)

Area of a Triangle in the xy -Plane

The area of a triangle with vertices

(x_1, y_1) , (x_2, y_2) , and (x_3, y_3)

is

$$\text{Area} = \pm \frac{1}{2} \det \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix}$$

where the sign ($\pm$) is chosen to give a positive area.

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Area, Volume, and Equations of Lines and Planes (2 of 6)

Test for Collinear Points in the xy -Plane

Three points (x_1, y_1) , (x_2, y_2) , and (x_3, y_3) are collinear if and only if

$$\det \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix} = 0.$$

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Area, Volume, and Equations of Lines and Planes (3 of 6)

Two-Point Form of an Equation of a Line

An equation of the line passing through the distinct points (x_1, y_1) and (x_2, y_2) is

$$\det \begin{bmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{bmatrix} = 0.$$

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Example 6 – Finding an Equation of the Line Passing Through Two Points

Find an equation of the line passing through the points $(2, 4)$ and $(-1, 3)$.

Solution:

Let $(x_1, y_1) = (2, 4)$ and $(x_2, y_2) = (-1, 3)$. Applying the determinant formula for an equation of a line produces

$$\begin{vmatrix} x & y & 1 \\ 2 & 4 & 1 \\ -1 & 3 & 1 \end{vmatrix} = 0.$$

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Example 6 – Solution

To evaluate this determinant, expand by cofactors in the first row.

$$x \begin{vmatrix} 4 & 1 \\ 3 & 1 \end{vmatrix} - y \begin{vmatrix} 2 & 1 \\ -1 & 1 \end{vmatrix} + 1 \begin{vmatrix} 2 & 4 \\ -1 & 3 \end{vmatrix} = 0$$

$$x(1) - y(3) + 1(10) = 0$$

$$x - 3y + 10 = 0$$

So, an equation of the line is $x - 3y = -10$.

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Area, Volume, and Equations of Lines and Planes (4 of 6)

Volume of a Tetrahedron

The volume of a tetrahedron with vertices (x_1, y_1, z_1) , (x_2, y_2, z_2) , (x_3, y_3, z_3) , and (x_4, y_4, z_4) is

$$\text{Volume} = \pm \frac{1}{6} \det \begin{bmatrix} x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \\ x_4 & y_4 & z_4 & 1 \end{bmatrix}$$

where the sign ($\pm$) is chosen to give a positive volume.

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Example 7 – Finding the Volume of a Tetrahedron

Find the volume of the tetrahedron shown in Figure 3.3.

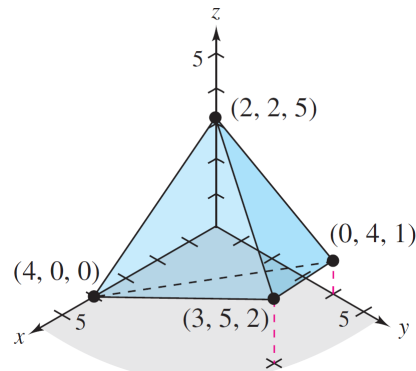


Figure 3.3

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Example 7 – Solution

Using the determinant formula for the volume of a tetrahedron produces

$$\frac{1}{6} \begin{vmatrix} 0 & 4 & 1 & 1 \\ 4 & 0 & 0 & 1 \\ 3 & 5 & 2 & 1 \\ 2 & 2 & 5 & 1 \end{vmatrix} = \frac{1}{6}(-72) = -12.$$

So, the volume of the tetrahedron is 12 cubic units.

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Area, Volume, and Equations of Lines and Planes (5 of 6)

Test for Coplanar Points in Space

Four points (x_1, y_1, z_1) , (x_2, y_2, z_2) , (x_3, y_3, z_3) , and (x_4, y_4, z_4) are coplanar if and only if

$$\det \begin{bmatrix} x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \\ x_4 & y_4 & z_4 & 1 \end{bmatrix} = 0.$$

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Area, Volume, and Equations of Lines and Planes (6 of 6)

Three-Point Form of an Equation of a Plane

An equation of the plane passing through the distinct points (x_1, y_1, z_1) , (x_2, y_2, z_2) , and (x_3, y_3, z_3) is

$$\det \begin{bmatrix} x & y & z & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \end{bmatrix} = 0.$$

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Example 8 – Finding an Equation of the Plane Passing Through Three points

Find an equation of the plane passing through the points $(0, 1, 0)$, $(-1, 3, 2)$, and $(-2, 0, 1)$.

Solution:

Using the determinant form of an equation of a plane produces

$$\begin{vmatrix} x & y & z & 1 \\ 0 & 1 & 0 & 1 \\ -1 & 3 & 2 & 1 \\ -2 & 0 & 1 & 1 \end{vmatrix} = 0.$$

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Example 8 – Solution (1 of 2)

To evaluate this determinant, subtract the fourth column from the second column to obtain

$$\begin{vmatrix} x & y-1 & z & 1 \\ 0 & 0 & 0 & 1 \\ -1 & 2 & 2 & 1 \\ -2 & -1 & 1 & 1 \end{vmatrix} = 0.$$

Expand by cofactors in the second row.

$$x \begin{vmatrix} 2 & 2 \\ -1 & 1 \end{vmatrix} - (y-1) \begin{vmatrix} -1 & 2 \\ -2 & 1 \end{vmatrix} + z \begin{vmatrix} -1 & 2 \\ -2 & -1 \end{vmatrix} = 0$$

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Example 8 – Solution (2 of 2)

$$x(4) - (y - 1)(3) + z(5) = 0$$

This produces the equation $4x - 3y + 5z = -3$.

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